10.1.1 Subspace iteration with a Hermitian matrix

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Week 10 due Mar 27, 2025 22:10 IST

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Start of transcript. Skip to the end.

We ended last week with the discussion of the Power Method, the Inverse Power

Method, and the Rayleigh Quotient Iteration. Now those methods allowed you

to find one eigenvector and one associated eigenvalue. In the

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Reading assignment

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Read Unit 10.1.1 of the notes. [LINK]

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Homework 10.1.1.1

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You may want to start by executing `git pull` to update your directory `Assignments`.

Examine Assignments/Week10/matlab/PowerMethod.m which implements

```
[ lambda_0, v0 ] = PowerMethod( A, x, maxiters, illustrate,
delay );
```

This routine implement the Power Method, starting with a vector `x` for a maximum number of iterations `maxiters` or until convergence, whichever comes first. To test it, execute the script in Assignments/Week10/matlab/test_SubspaceIteration.m which uses the Power Method to compute the largest eigenvalue (in magnitude) and corresponding eigenvector for an $m \times m$ Hermitian matrix A with eigenvalues $1, \ldots, m$.

Be sure to click on "Figure 1" to see the graph that is created.

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we could say, "Hmm, we'll run the power method." We will find an eigenvector

associated with the largest eigenvalue. We will call that vector that we find x_0.

When we pick a random vector, we start by subtracting out the component of v in

the direction of x_0, which is equal to -- let's see -- v minus x_0 transpose v times

x_0. Well, if x_0 of course had been normalized to have length 1. And then

once we have picked

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Homework 10.1.1.2

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Copy Assignments/Week10/matlab/PowerMethod.m into `PowerMethodLambda1.m` . Modify it by adding an input parameter `x0` , which is an eigenvector associated with λ_0 (the eigenvalue with largest magnitude).

```
[ lambda_1, v1 ] = PowerMethodLambda1( A, x, v0, maxiters,
illustrate, delay )
```

The new function should subtract this vector from the initial random vector as in the above algorithm.

Modify the appropriate line in Assignments/Week10/matlab/test_SubspaceIteration.m>, changing (0) to (1), and use it to examine the convergence of the method.

What do you observe?

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in on when we were not subtracting out the

component in the direction of x_0. Alright? So how can we then force that

component out every time. Well, I guess I kind of gave it away. We can force that

component out every time. We can say, "Hmm, before we normalize this vector let's go

ahead and be diligent and subtract out again any component in the direction of

x_0 that has snuck in." Oops I should have had Hermitian

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Homework 10.1.1.3

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Copy `PowerMethodLambda1.m` into `Assignments/Week10/matlab/PowerMethodLambda1Reorth.m` and modify it to reorthogonalize with respect to `x0` :

```
[ lambda_1, v1 ] = PowerMethodLambda1Reorth( A, x, v0, maxiters,
illustrate, delay  );
```

You can test your implementation by commenting out the obvious statement(s) in Assignments/Week10/matlab/test_SubspaceIteration.m, changing `(0)` to `(1)`, and use it to examine the convergence of the method.

What do you observe?

- [x] done

Solution:

For a complete solution, see the notes .

There is a video in the notes.

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Answers are displayed within the problem

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ignoring the fact that this also returns R because we're never even

going to use that matrix R. But you know for, maybe for completeness I should say

we get the matrix v_0 ^(0) v_1^(0) comma R equal to that because

typically our routine for computing the QR factorization of a matrix also

returns our matrix R. Then down here what we could do is say, "Hmm, let's create

temporary vectors v_0 v_1 by

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vectors to be of length one, so we don't need to divide. And then like the one you

would get by looking at v_1^(k) Hermitian A v_1^(k). And one elegant way of

kind of looking at that simultaneously is to say, "Well, if we look at the matrix

that comes from doing v_0^(k) v_1^(k) transpose times A times the v_0^(k) v_1^(k)

like that, what you expect is that that matrix, it's obviously a 2 by 2 matrix,

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Homework 10.1.1.4

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